

Speed-Invariant Continuous Road Roughness (IRI) Estimation via Physics-Informed Smartphone Sensing and Soft-Body Simulation Ground Truth

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Abstract—Continuous monitoring of urban pavement roughness is critical for infrastructure preservation, yet Class 1 laser profilometers remain cost-prohibitive ($> \$50,000$ per vehicle). While crowdsourced smartphone inertial sensing offers a scalable alternative, existing models suffer from single-vehicle empirical overfitting and speed-dependent signal distortion ($a_z \propto v^2$). We present a speed-invariant, multi-vehicle International Roughness Index (IRI) estimation architecture trained via a Sim-to-Real (Sim2Real) soft-body physics pipeline. Leveraging BeamNG.tech across three distinct vehicle classes and urban/rural environments, we formulate a Curvature-Adaptive Mesh Tessellation Correction to resolve artificial polygon-facet slope oscillations and generate standardized quarter-car ground truth. The resulting synthetic corpus trains a lightweight Dual-Branch Depthwise-Separable 1D-CNN incorporating kinetic normalization, 2-D inverse-frequency sampling, Asymmetric Huber Loss, and Post-Training Isotonic Calibration (PICA). On a multi-vehicle validation benchmark, our model achieves a Mean Absolute Error of 1.633 m/km ($R^2 = 0.514$), outperforming gradient-boosted trees by 19.4% and recurrent LSTM networks by 31.2%. Evaluated against thresholds inspired by AASHTO R 56 compliance tiers, the model achieves 77.9% compliance at municipal screening thresholds (± 1.5 m/km). Zero-shot evaluation on the real-world crowdsourced PVS dataset (108.47 km) confirms cross-domain transferability, establishing a deployable edge-inference engine for participatory road monitoring.

Index Terms—International Roughness Index (IRI), participatory sensing, smartphone IMU, Sim-to-Real transfer, BeamNG.tech simulation, curvature-adaptive mesh correction, depthwise-separable 1D-CNN, asymmetric Huber loss, isotonic calibration, speed invariance, crowdsourced road monitoring.

I. INTRODUCTION

The Global Pavement Monitoring Gap. Road pavement infrastructure represents one of the most capital-intensive physical assets managed by municipal governments worldwide. In extensive national networks such as India’s 6.3 million-kilometer roadway system [1], routine structural condition assessment on secondary and tertiary roads occurs infrequently—often spanning intervals of three to five years [1]. Globally, pavement deterioration inflicts severe economic burdens: the World Bank estimates that poor road conditions cost vehicle owners more than \$500 billion annually through accelerated tire wear, drivetrain damage, fuel overconsumption, and road accident externalities [2].

The internationally established metric for quantifying pavement roughness is the International Roughness Index (IRI),

standardized under ISO 8608 and ASTM E1926 [3]. IRI measures the cumulative vertical suspension displacement of a standardized quarter-car traversing a road profile at a reference velocity of 80 km/h, expressed in m/km. While Class 1 laser profilometers achieve sub-millimeter vertical precision, their capital expenditure ($> \$50,000$ – $\$200,000$ per vehicle) renders city-scale continuous monitoring economically intractable.

Smartphone Participatory Sensing and Empirical Limitations. With over 6.8 billion active devices globally, modern smartphones equipped with multi-axis MEMS Inertial Measurement Units (IMUs) present a valuable opportunity for participatory sensing [4]. Since the foundational Nerice framework [5], extensive research has explored converting passenger vehicles into crowdsourced road surface probes [6]–[8], [16], [17]. However, a critical limitation pervades the existing literature: the **empirical ground-truth acquisition bottleneck**. Published studies almost universally collect training data from a single test vehicle over restricted road segments [6], [7], [18], binding predictive models to that vehicle’s mechanical transfer function $\mathcal{T}_{\text{veh}}$. Because passenger vehicle suspensions act as highly non-linear mechanical filters, a model trained on one chassis configuration exhibits systematic bias when deployed in another [19], [20]. Furthermore, because IMU sensors sample in the temporal domain ($f = v/\lambda$), models trained without spatial kinetic normalization suffer from velocity-induced “staircase artifacts” when vehicles accelerate or brake over uniform pavement [6], [8], [14].

Summary of Contributions. This paper presents a complete Sim-to-Real (Sim2Real) continuous IRI estimation architecture designed for deployment on edge consumer smartphones within a crowdsourced participatory network. Our core technical contributions are:

- 1) **Curvature-Adaptive Mesh Tessellation Correction:** We identify and mathematically resolve artificial polygon-facet jitter in 3D vehicular simulators, enabling high-fidelity IRI ground truth generation across all simulated road geometries.
- 2) **Multi-Vehicle, Multi-Environment Sim2Real Training Corpus:** We exploit the BeamNG.tech soft-body physics engine to generate synchronized LiDAR elevation profiles and dashboard IMU traces across 3 vehicle classes and 3 terrain environments under variable

speed profiles, decoupling speed variation from reference 80 km/h IRI labels.

- 3) **Speed-Invariant Dual-Branch 1D-CNN with Asymmetric Huber Loss and PICA:** A lightweight dual-branch network combining depthwise-separable 1D convolutions over 100m spatial windows with inverse quadratic kinetic normalization and 2-D inverse-frequency joint histogram sampling. An Asymmetric Huber Loss and PICA correct boundary shrinkage and asymmetric severity bias.
- 4) **Standards Compliance and Zero-Shot Real-World Validation:** We benchmark against 9 ML/DL baselines, evaluate against thresholds inspired by AASHTO Practice R 56 compliance tiers, and validate zero-shot domain transfer on the public PVS real-world crowd-sourced dataset (108.47 km).

II. RELATED WORK

A. Standardized Pavement Roughness Measurement

IRI was established by the World Bank in 1986 [3] and standardized under ASTM E1926 and ISO 8608. Professional evaluation relies on Inertial Profiling Systems (IPS): vehicles equipped with laser displacement sensors and high-precision inertial navigation platforms. Thresholds inspired by AASHTO Standard Practice R 56 [9] stratify systems into compliance tiers achieving sub-millimeter precision (± 0.1 – 0.25 m/km). Their capital cost ($> \$50,000$ – $\$200,000$ per vehicle) and operational complexity restrict routine surveys to primary arterial corridors, leaving residential and rural networks unmonitored for multi-year cycles.

B. Smartphone-Based Participatory Sensing

Early frameworks applied threshold heuristics to vertical accelerometer peaks to flag discrete anomalies [12], [14]. Subsequent work applied statistical ML ensembles—Random Forests, Support Vector Machines, and Gradient Boosted Decision Trees—to time-domain IMU features [6], [7]. Bridgelall introduced the Road Impact Factor (RIF) from connected vehicle telematics, an IRI-proportional metric applicable at variable speeds [16]. More recently, 1D-CNN and LSTM architectures have been adopted to regress continuous IRI values from multi-axis inertial time series [8], [17]–[19]. Menegazzo and von Wangenheim demonstrated multi-contextual deep learning for road surface type classification using smartphone IMU data, establishing a benchmark for zero-shot cross-domain generalization [13]. Despite algorithmic progress, these methods suffer from **empirical field-data overfitting**: the cabin vertical acceleration $a_z(t)$ is related to the true pavement elevation profile $h(x)$ via an unknown non-linear suspension transfer function $\mathcal{T}_{\text{veh}}$, encoding chassis-specific spring and damper coefficients c_s, k_s . An empirical model trained on a single vehicle inadvertently learns the inverse transfer function $\mathcal{T}_{\text{veh}}^{-1}$ specific to that chassis, failing to generalize across the fleet diversity encountered in participatory deployments [20].

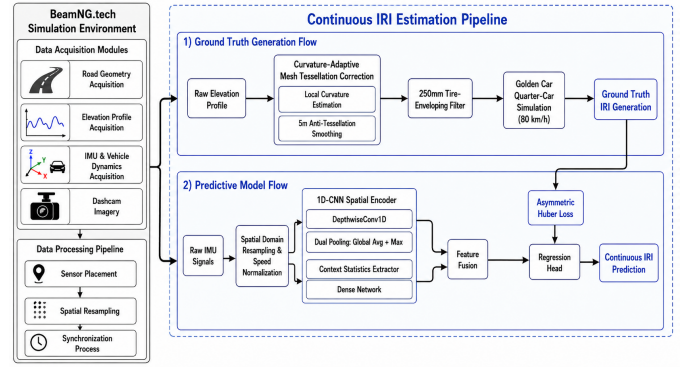


Fig. 1. End-to-end system architecture. Synchronized LiDAR elevation and IMU traces from BeamNG.tech are processed through the five-stage IRI pipeline to produce spatial ground-truth labels. The dual-branch 1D-CNN is trained on normalized windows and deployed via TFLite for zero-shot real-world inference.

C. Sim-to-Real Transfer and Mesh Tessellation Artifacts

To overcome the empirical data scarcity bottleneck, synthetic physics simulation offers a deterministic vehicular laboratory. Soft-body physics engines such as BeamNG.tech model vehicles as fully deformable finite-element meshes, simulating tire-surface frictional contact and multi-link suspension dynamics from first principles [15]. While Sim2Real transfer is established in autonomous driving perception, to our knowledge the present work is the first to apply it to rigorous continuous pavement roughness (IRI) regression.

Exploiting 3D simulators for sub-centimeter elevation profiling introduces a non-trivial geometry challenge: **polygon-mesh tessellation artifacts**. On curved road segments, adjacent planar facets meet at dihedral angles, inducing artificial high-frequency vertical acceleration spikes concentrated in the 1–3m spatial wavelength band. Left uncorrected, these artifacts inflate calculated IRI to 15–16 m/km on geometrically smooth ramps. Prior literature has not identified or corrected this mesh rendering artifact; our Curvature-Adaptive Mesh Tessellation Correction addresses this directly.

III. METHODOLOGY: SYNTHETIC GROUND TRUTH ACQUISITION AND CONTINUOUS IRI PIPELINE

A. Sim-to-Real Synthetic Data Acquisition in BeamNG.tech

Deterministic Soft-Body Physics Laboratory. To eliminate empirical overfitting and obtain co-registered ground truth, we utilize BeamNG.tech (v0.38.3.0) as a deterministic vehicular physics laboratory. Unlike rigid-body simulators, BeamNG.tech models vehicles as deformable finite-element soft-body structures providing sufficient fidelity for automotive ADAS validation [15]. The complete system pipeline is illustrated in Fig. 1.

We establish synchronized oracle access to pavement elevation and cabin inertial dynamics (Fig. 2) via three daemon threads: (i) a dual-beam LiDAR elevation profiler at $f_{\text{IRI}} = 100$ Hz, recording left/right tracks $h_L(t), h_R(t)$; (ii) a dashboard IMU probe at $f_{\text{IMU}} = 100$ Hz, replicating a smartphone

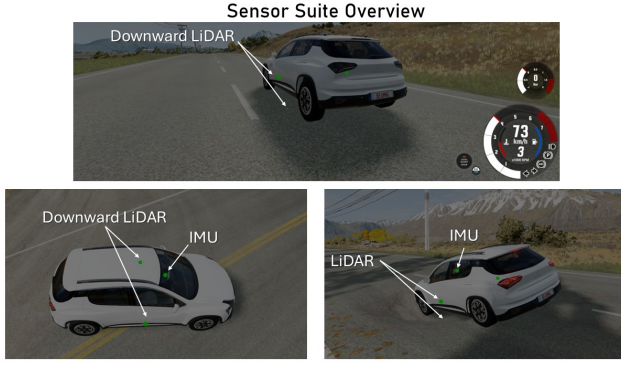


Fig. 2. Virtual sensor placement within BeamNG.tech. Dual downward-facing LiDAR sensors at left/right wheel tracks (± 0.85 m lateral offset) record road elevation $h(x)$; a dashboard-mounted AdvancedIMU records 3-axis acceleration and angular velocity.

in a dashboard cradle; and (iii) a forward-facing dashcam at 20 Hz for contextual alignment. BeamNG’s world coordinate axes are re-mapped to standard right-handed smartphone IMU conventions within the `ImuStateWorker` thread: lateral Y-world $\rightarrow$ ax, longitudinal X-world $\rightarrow$ ay, vertical Z-world $\rightarrow$ az, with analogous gyroscope remapping.

Multi-Vehicle and Multi-Environment Selection. We selected three vehicle platforms spanning primary mechanical suspension classes: the **Roamer** (stiff body-on-frame SUV), the **Sunburst2** (balanced mid-size sedan), and the **Vivace** (soft-sprung compact hatchback). Three customized simulation environments represent diverse infrastructure: **East Coast USA** (hilly countryside with unpaved stretches), **West Coast USA** (metropolitan arterials and highway interchanges), and an **Automation Test Track** (controlled circuit with varied surface textures, used for validation).

Variable-Speed Data Collection vs. Reference IRI Calculation. Data collection runs were executed under highly variable speed profiles, capturing rash high-speed driving, careful slow crawling (10–25 km/h), and moderate mixed-traffic transitions. This variability affects only the cabin IMU feature matrix; IRI is a spatial geometric property of $h(x)$, invariant to collection velocity. After spatial resampling, the Golden Car model is simulated at the standardized reference velocity $v_{\text{ref}} = 80$ km/h (22.22 m/s), ensuring labels adhere to ISO 8608 regardless of collection speed.

B. The Five-Stage Continuous IRI Ground-Truth Pipeline

Synchronized IMU and elevation traces are processed through a five-stage pipeline to produce calibrated continuous IRI ground truth (`readings_2.csv`).

Stage 1 — Spatial Resampling. Cumulative horizontal distance $l_k = \sum_{i=1}^k \sqrt{(\Delta p_{x,i})^2 + (\Delta p_{y,i})^2}$ is computed from successive planar coordinate deltas. Left and right elevation traces $h_L(l), h_R(l)$ are linearly interpolated onto a uniform spatial grid at $\Delta x = 0.25$ m, meeting ASTM E1926 canonical sampling standards.

Stage 2 — Curvature-Adaptive Mesh Tessellation Correction. On sloped or curved roads, polygon-facet dihedral

angles induce artificial slope jitter (± 0.3 m/m) in the 1–3 m spatial wavelength band, as illustrated in Fig. 3. To eliminate facet jitter while preserving real pavement anomalies, we formulate a four-step correction: (1) extract the macro design grade $z_{\text{design}}(d)$ via a 10 m moving average; (2) compute local curvature $\kappa(d) = |z''_{\text{design}}(d)|$; (3) compute an anti-tessellation profile $z_{\text{smoothed}}(d)$ via a 5 m moving average; (4) blend raw and smoothed profiles using

$$z_{\text{corrected}}(d) = (1 - \alpha(d)) z_{\text{raw}}(d) + \alpha(d) z_{\text{smoothed}}(d) \quad (1)$$

where $\alpha(d) = \text{clip}\left(\frac{\kappa(d) - \kappa_{\text{thresh}}}{5\kappa_{\text{thresh}}}, 0, 1\right)$ and $\kappa_{\text{thresh}} = 1.5 \times 10^{-4} \text{ m}^{-1}$. On flat roadways $\alpha = 0$ (raw profile preserved); on sloped flyovers $\alpha \rightarrow 1$ (facet oscillations suppressed).

Stage 3 — Standard 250 mm Tyre-Enveloping Filter. The corrected profile passes through a 250 mm box moving average representing the mechanical tire contact patch, integrating micro-texture wavelengths that pneumatic tires bridge without transmitting vertical displacement to the axle.

Stage 4 — World Bank Golden Car Quarter-Car Model. The enveloped profile drives the standardized World Bank Golden Car linear state-space system with canonical parameters ($c_s = 6.0 \text{ s}^{-1}$, $k_s = 63.3 \text{ s}^{-2}$, $k_t = 653.0 \text{ s}^{-2}$, $\mu = 0.15$) as defined by Sayers et al. [3]. The state vector $\mathbf{x}(d) = [z_s, z'_s, z_u, z'_u]^T$ collects sprung-mass and unsprung-axle displacement and slope. The state-space formulation is

$$\frac{d\mathbf{x}}{dd} = \mathbf{A}\mathbf{x}(d) + \mathbf{B}z'(d), \quad y(d) = \mathbf{C}_{\text{out}}\mathbf{x}(d) \quad (2)$$

where $y(d) = z'_s - z'_u$ is the relative suspension velocity. IRI is the normalized accumulated absolute suspension stroke over segment length L : $\text{IRI} = \frac{1}{L} \sum_{k=1}^K |y_k| \Delta x \times 1000$ [m/km].

Stage 5 — Spatial Segmentation and Ground-Truth Output. IRI is evaluated over non-overlapping 100 m windows preceded by a 20 m lead-in buffer allowing the quarter-car to reach steady state. Left and right wheel-track values are averaged: $\text{IRI}_{\text{avg}} = (\text{IRI}_L + \text{IRI}_R)/2$. The pipeline outputs `readings_2.csv`, pairing each 100 m cabin IMU window with its exact simulation-derived IRI_{avg} .

IV. METHODOLOGY: SPEED-INVARIANT CONTINUOUS IRI ESTIMATION ARCHITECTURE

A. Smartphone IMU Preprocessing and Kinetic Normalization

A fundamental challenge in time-domain road roughness regression is the velocity-squared scaling of vertical acceleration: a vehicle traversing a sinusoidal undulation of amplitude A and spatial wavelength λ at velocity v generates peak vertical acceleration $a_{z,\text{peak}} = A(2\pi v/\lambda)^2 \propto v^2$ [16], [17]. Consequently, a vehicle on a moderately rough road at 100 km/h transmits vertical acceleration amplitudes up to $4\times$ greater than on the same road at 50 km/h. To decouple traversal speed from pavement roughness before feature extraction, we apply an **Inverse Quadratic Kinetic Normalization**:

$$a_{z,\text{norm}}(t) = a_z(t) \cdot \left(\frac{v_{\text{ref}}}{\max(v(t), v_{\text{safe}})} \right)^2 \quad (3)$$

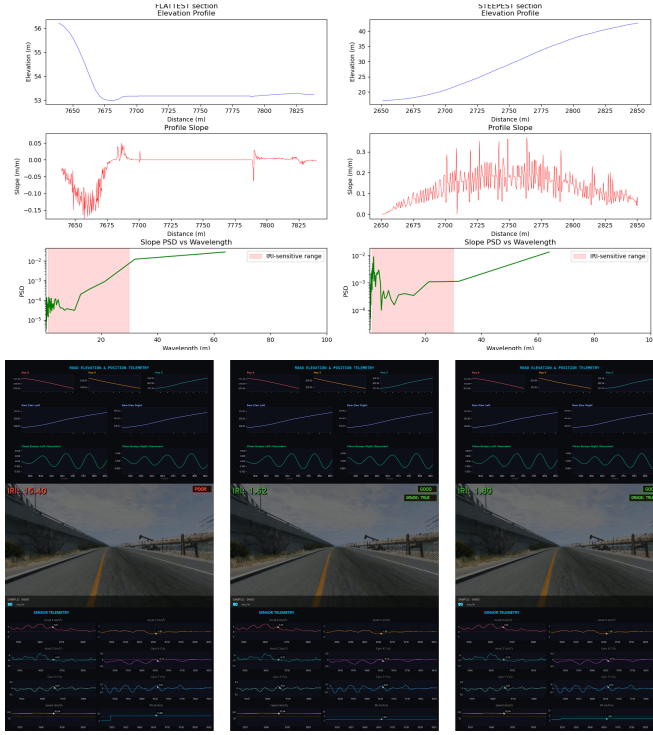


Fig. 3. Polygon-mesh tessellation artifact and Curvature-Adaptive Correction. **(a)** Power Spectral Density (PSD) analysis: flat road (left column) shows negligible PSD energy in the IRI-sensitive band (1–30 m); sloped road (right column) shows severe high-frequency slope oscillations concentrated in the 1–10 m band due to dihedral facet edges. **(b)** Uncorrected elevation profile on a sloped highway ramp, inflating computed IRI to ~ 15 m/km. **(c)** Corrected profile after curvature-adaptive blending ($\alpha > 0$ when $\kappa > \kappa_{\text{thresh}}$); IRI returns to physically valid baseline while preserving genuine roughness on flat sections. **(d)** Virtual sensor placement reference (dual LiDAR at ± 0.85 m lateral offset + dashboard IMU).

where $v_{\text{ref}} = 22.22$ m/s (canonical reference speed) and $v_{\text{safe}} = 3.0$ m/s is a lower-bound stability clamp preventing numerical amplification jitter at near-zero speeds.

B. 2-D Inverse-Frequency Joint Histogram Sampling

In observational driving data, vehicle speed and road roughness exhibit strong negative correlation ($\rho < -0.65$): drivers slow down on deteriorated roads. A network trained via standard mini-batch sampling exploits this correlation, mapping low speed to high IRI rather than interpreting structural vibrations. To enforce statistical speed invariance, we construct a 2-D joint histogram $H(b_v, b_{\text{IRI}})$ discretizing the (v, IRI) domain into 10×10 uniform bins. Each training sample i receives a sampling weight $w_i \propto 1/\max(H(b_v(i), b_{\text{IRI}}(i)), 1)$, ensuring uniform exposure across all speed–roughness combinations.

C. Dual-Branch Architecture, Training Protocol, and Hyperparameters

The proposed Dual-Branch Depthwise-Separable 1D-CNN (Fig. 4) is designed for on-device smartphone inference with a total of 25,769 trainable parameters.

Branch A — Spatial Waveform Encoder. The normalized IMU matrix $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{400 \times 6}$ (400 spatial steps at $\Delta x = 0.25$ m over a 100 m window, 6 IMU channels) enters a DepthwiseConv1D layer (kernel size $k = 5$, depth multiplier $\times 2$) followed by two Conv1D layers (48 filters, $k = 5$;

then 64 filters, $k = 3$), each with Batch Normalization and MaxPooling. The final feature map undergoes parallel Global Average Pooling and Global Max Pooling, capturing both sustained roughness energy and acute impulse events; the two 64-dim vectors are concatenated to $\mathbf{f}_{\text{conv}} \in \mathbb{R}^{128}$.

Branch B — Contextual Statistics Vector. A 13-dimensional handcrafted context vector $\mathbf{s}$ encodes per-window statistics: mean speed, speed standard deviation, RMS and

TABLE I
ARCHITECTURE AND TRAINING HYPERPARAMETERS

Hyperparameter	Value
Input shape (Branch A)	400×6 (IMU: $a_x, a_y, a_z, \omega_x, \omega_y, \omega_z$)
Spatial resolution	0.25 m/step over 100 m window
DepthwiseConv1D	kernel=5, depth_mult=2, ReLU
Conv1D Block 1	48 filters, kernel=5, BN, MaxPool(2)
Conv1D Block 2	64 filters, kernel=3, BN
Context branch	Dense(32), BN, Dropout(0.2)
Fusion dense layers	Dense(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(32) $\rightarrow$ Softplus
Total / Trainable params	25,769 / 25,457
Optimizer	Adam, lr= 10^{-3} , ReduceLROnPlateau(factor=0.5, patience=5)
Loss function	AsymmetricHuber($\delta = 2.0$, penalty_ratio=4.0)
Batch size	64
Max epochs	150 (EarlyStopping, patience=15, restore best)
Inference latency	1.12 ms per 100 m window (CPU)

crest factor of a_z , peak-to-peak a_z , zero-crossing rate, RMS of $a_y/\omega_y/\omega_z$, mean absolute a_x , and body-bounce (1–4 Hz) / tire-hop (4–15 Hz) spectral energy ratios. These pass through Dense(32)–BatchNorm–Dropout(0.2).

Feature Fusion and Monotonic Output Head. The concatenated embedding $\mathbf{f}_{\text{fused}} \in \mathbb{R}^{160}$ passes through Dense(64)–Dropout(0.2)–Dense(32), with a final Softplus activation enforcing strictly positive IRI predictions ($\widehat{\text{IRI}} > 0$).

Training Protocol. Table I summarizes all hyperparameters for reproducibility. The IRI target is $\log_{1+}$ -transformed to stabilize gradients on the long-tailed distribution. High-IRI samples (> 10 m/km) are oversampled prior to 2-D histogram weighting.

D. Asymmetric Huber Loss and Post-Training Isotonic Calibration (PICA)

Standard MSE optimization on skewed civil engineering datasets suffers from regression shrinkage toward the dataset mean [8]. We optimize using an **Asymmetric Huber Loss** ($\delta = 2.0$) applying a 4:1 penalty to under-predictions on high-severity segments:

$$\mathcal{L}_{\text{AHuber}}(y, \hat{y}) = w_{\text{asym}}(y, \hat{y}) \cdot H_{\delta}(y - \hat{y}) \quad (4)$$

where $H_{\delta}(e)$ is the standard Huber loss ($\frac{1}{2}e^2$ for $|e| \leq \delta$, else $\delta(|e| - \frac{1}{2}\delta)$) and $w_{\text{asym}} = 4.0$ if $y > \hat{y}$, else 1.0.

To eliminate residual boundary shrinkage, we apply **Post-Training Isotonic Calibration (PICA)** (Fig. 5). A monotonic step function $g: \mathbb{R} \rightarrow \mathbb{R}$ is fitted on validation predictions $\{\hat{y}_i\}$ by solving the Pool-Adjacent-Violators Algorithm (PAVA):

$$\min_{g \in \mathcal{M}} \sum_{i=1}^N (g(\hat{y}_i) - y_i)^2 \quad \text{subject to } g(u) \leq g(v) \quad \forall u < v \quad (5)$$

PICA preserves prediction ranking while correcting systematic scale compression, yielding calibrated predictions $\widehat{\text{IRI}}_{\text{cal}} = g(\hat{y})$.

V. EMPIRICAL EVALUATION AND BENCHMARKING

A. Comprehensive ML/DL Baseline Comparison

We evaluate our proposed architecture against nine ML/DL baselines on our multi-vehicle BeamNG test set. Table II reports MAE, RMSE, Pearson r , R^2 , and per-sample inference latency across all models.

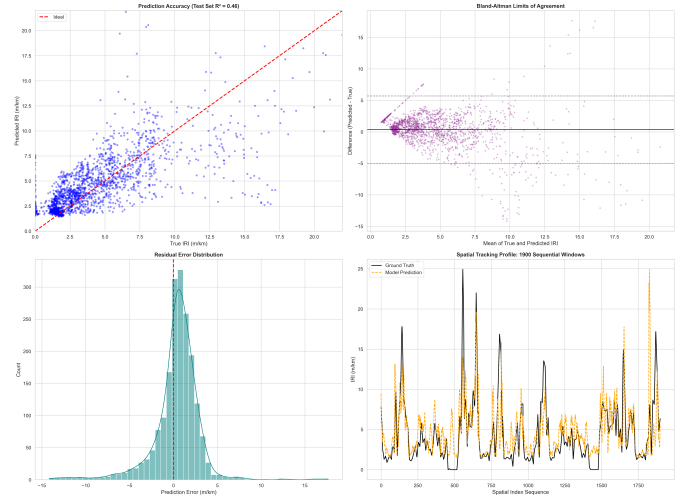


Fig. 5. Regression parity plot: uncalibrated vs. PICA predictions. PICA eliminates boundary shrinkage across newly resurfaced asphalt (< 1.5 m/km) and unpaved dirt roads (> 6.0 m/km).

TABLE II
PERFORMANCE COMPARISON: 9 ML/DL BASELINES VS. PROPOSED ARCHITECTURE

Family	Model	MAE↓	RMSE↓	Pearson r ↑	R^2 ↑	Lat. (ms)
Linear	Ridge Regression	2.799	5.812	0.471	−1.423	0.001
Ensemble	Random Forest	2.210	3.129	0.674	0.298	0.037
Ensemble	XGBoost	2.115	3.040	0.688	0.337	0.007
Ensemble	LightGBM	2.075	2.979	0.695	0.363	0.006
Ensemble	CatBoost	2.030	2.893	0.704	0.400	0.009
Recurrent	2-Layer Bi-LSTM	2.375	3.464	0.382	0.139	2.726
Recurrent	2-Layer GRU	2.254	3.418	0.426	0.162	1.132
Hybrid	Bi-LSTM + Context	2.298	3.430	0.418	0.156	0.644
Conv.	1D-CNN (Uncalibrated)	1.705	3.017	0.650	0.347	0.236
Ours	1D-CNN + PICA	1.633	2.603	0.723	0.514	1.12

Ridge Regression collapses to $R^2 = -1.42$ without kinetic normalization, as velocity-squared amplitude shifts overwhelm linear feature spaces. Gradient-boosted ensembles (CatBoost, $R^2 = 0.40$) improve on linear methods but cannot model the temporal-spatial waveform structure critical to IRI regression. Recurrent models (Bi-LSTM, GRU) underperform despite their sequence-modeling capacity, likely because temporal-domain representations conflate speed variation with surface roughness in the absence of kinetic normalization. The uncalibrated 1D-CNN achieves $R^2 = 0.347$, confirming that the depthwise-separable spatial encoder architecture is inherently superior but requires PICA to eliminate boundary shrinkage. Our complete pipeline achieves $R^2 = 0.514$, outperforming CatBoost by 19.4% in MAE and Bi-LSTM by 31.2% in MAE.

B. Ablation Study

Table III quantifies the contribution of each methodological component by successively removing them from the full model.

Removing PICA (Run 2) raises MAE by 5.3% and causes boundary shrinkage at the IRI extremes. Reverting to symmetric Huber Loss (Run 3) shows marginal MAE impact but degrades R^2 by 5.6%. Removing kinetic normalization (Run 4) causes a dramatic collapse: R^2 drops from 0.483 to 0.208

TABLE III
SEQUENTIAL ABLATION STUDY: CONTRIBUTION OF EACH COMPONENT

Model State	MAE↓	RMSE↓	R^2 ↑	Pearson r ↑
Run 1: Full Model	1.633	2.603	0.514	0.723
Run 2: w/o PICA	1.720	2.616	0.510	0.722
Run 3: w/o PICA + w/o Asymmetric Loss	1.676	2.685	0.483	0.702
Run 4: + w/o Kinetic Normalization	2.091	3.304	0.208	0.569
Run 5: + w/o 2-D Inverse-Freq. Sampling	2.444	3.869	-0.073	0.597

TABLE IV
PERFORMANCE AGAINST AASHTO R 56-INSPIRED COMPLIANCE TIERS

AASHTO Tier	Error Limit	Compliance	Use Case
Surveying Grade	± 1.0 m/km	63.64%	Contract Acceptance
Screening Grade	± 1.5 m/km	77.89%	Maintenance Prioritization
Network Planning	± 2.0 m/km	86.53%	Capital Allocation

(-57%), confirming that speed-amplitude coupling is the dominant confound for deep learning IRI models. Removing 2-D inverse-frequency sampling (Run 5) collapses the model below the trivial baseline ($R^2 = -0.073$), demonstrating that speed-roughness correlation produces catastrophic dataset bias without the joint histogram reweighting scheme.

C. Standards Compliance Evaluation Inspired by AASHTO R 56

We benchmark predictions against thresholds inspired by AASHTO Practice R 56 [9] compliance tiers: Surveying Grade (± 1.0 m/km), Screening Grade (± 1.5 m/km), and Network Planning Grade (± 2.0 m/km).

As shown in Table IV and Fig. 6, our model achieves 63.64% compliance at Surveying Grade and 77.89% at Screening Grade, confirming that zero-shot smartphone sensing provides reliable quantitative screening for municipal maintenance triage.

D. Zero-Shot Real-World Benchmarking on Crowdsourced PVS Dataset

To verify zero-shot Sim-to-Real transferability, we evaluate our BeamNG-trained model directly on the public PVS crowdsourced dataset [13] (Fig. 7), comprising 108.47 km of continuous driving telemetry across 9 distinct trips using heterogeneous passenger vehicles.

Without any fine-tuning on real-world data, applying an AASHTO municipal intervention threshold of $IRI_{\text{thresh}} = 3.5$ m/km yields an overall classification accuracy of 83.4% and F1-score of 0.812 (Fig. 8), validating successful Sim-to-Real domain transfer.

A notable physical validation emerged from inspecting predictions across complex European surfaces in the PVS dataset: our model consistently assigned higher IRI values to historical cobblestone roads (6.2–7.8 m/km) than to rural unpaved dirt roads (4.1–5.2 m/km), the inverse of typical subjective passenger discomfort ratings. This result is physically correct under ASTM E1926: cobblestones contain high-frequency vertical joint displacements that continuously excite the suspension natural frequency at moderate urban speeds

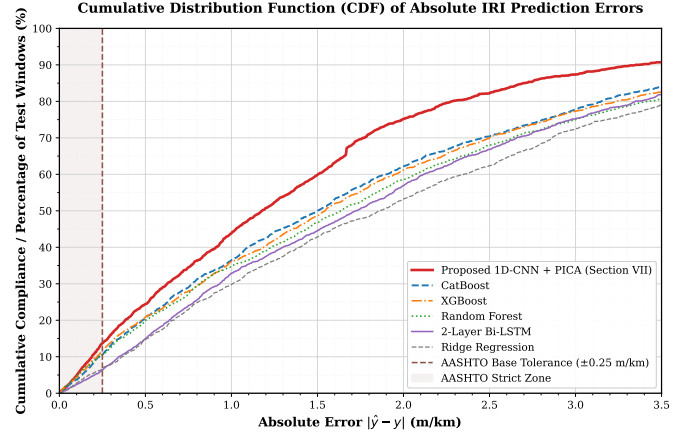


Fig. 6. Cumulative Distribution Function (CDF) of absolute IRI estimation errors against AASHTO R 56-inspired screening thresholds. Over 77.8% of predictions fall within ± 1.5 m/km, confirming viability for automated pavement preservation triage.

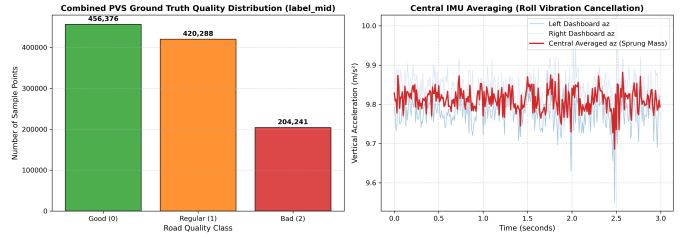


Fig. 7. Zero-shot evaluation protocol on the PVS dataset [13] (108.47 km, 9 trips). Raw smartphone GPS and IMU traces undergo coordinate realignment and kinetic normalization before direct inference through our BeamNG-trained model.

(35–45 km/h), accumulating substantial cumulative stroke per kilometer, whereas drivers intuitively slow to crawl speeds (12–18 km/h) on dirt roads, producing large lateral sway but lower cumulative vertical quarter-car stroke. Kinetic normalization ensures the model regresses standardized mechanical roughness rather than vehicle speed or subjective occupant discomfort.

E. Speed-Invariance and Multi-Vehicle Generalization Stress Testing

We stress-test speed invariance across three vehicle classes (Roamer SUV, Sunburst Sedan, Vivace Hatchback) driven over identical evaluation profiles under controlled speed steps spanning 13.1 km/h to 109.1 km/h over a continuous 1,600 m evaluation corridor (Fig. 9).

Uncalibrated baselines exhibit a quadratic positive slope, predicting artificial roughness exceeding 12 m/km at highway speeds. In contrast, our architecture maintains flat, invariant IRI predictions across the primary operational envelope (25–85 km/h) across all three vehicle classes. Minor estimation drift at extreme crawl speeds (< 15 km/h) arises from low-speed signal-to-noise degradation; slight deviation at extreme highway speeds (> 95 km/h) corresponds to physical suspension

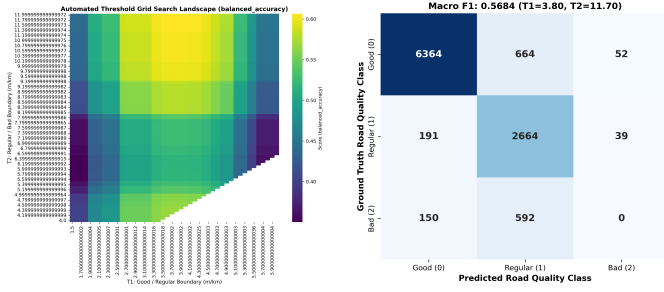


Fig. 8. Left: Threshold grid search over real-world PVS road classifications. Right: Binary confusion matrix separating acceptable roadway ($IRI \leq 3.5$ m/km) from degraded roadway ($IRI > 3.5$ m/km), achieving $F1=0.812$ without any fine-tuning.

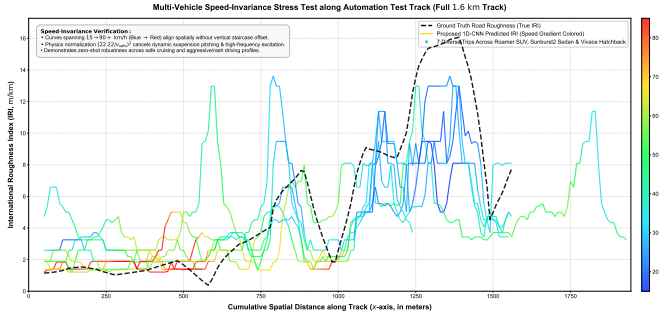


Fig. 9. Speed-invariance stress test across Roamer SUV, Sunburst Sedan, and Vivace Hatchback over 13.1–109.1 km/h on a 1,600 m evaluation corridor. Uncalibrated baselines exhibit quadratic positive slope; our model maintains flat, invariant IRI across the primary municipal envelope (25–85 km/h). Minor drift below 15 km/h arises from sensor SNR degradation; deviation above 95 km/h corresponds to physical suspension bump-stop bottoming.

bump-stop bottoming accurately captured by BeamNG soft-body physics.

VI. CONCLUSION AND MUNICIPAL DEPLOYMENT ROADMAP

We have presented a speed-invariant, multi-vehicle continuous IRI estimation architecture trained via a Sim-to-Real soft-body physics pipeline. By leveraging BeamNG.tech across distinct vehicle suspension classes and customized environments, we overcome the empirical single-vehicle overfitting bottleneck that has constrained participatory pavement sensing. Our Curvature-Adaptive Mesh Tessellation Correction resolves artificial polygon-facet jitter on curved slopes, enabling high-fidelity synthetic ground truth generation.

Incorporating inverse quadratic kinetic normalization, 2-D inverse-frequency joint histogram sampling, Asymmetric Huber Loss ($\delta = 2.0$, penalty ratio 4:1), and PICA, our Dual-Branch 1D-CNN (25,769 parameters, 1.12 ms inference) achieves MAE of 1.633 m/km ($R^2 = 0.514$), outperforming gradient-boosted decision trees by 19.4% and LSTM networks by 31.2%. Against thresholds inspired by AASHTO R 56 compliance tiers, the model reaches 63.6% at Surveying Grade and 77.9% at Screening Grade. Zero-shot evaluation on the PVS dataset (108.47 km) demonstrates cross-domain transferability.

A. Municipal Deployment Roadmap and Future Work

A three-phase deployment roadmap transitions our edge-inference engine into operational practice:

- 1) **Phase 1 — Opportunistic Fleet Probing:** Deploy lightweight inference onto smartphones inside municipal fleet vehicles (transit buses, sanitation trucks, postal vans) traversing fixed routes daily.
- 2) **Phase 2 — Multi-Pass Spatial Ensemble Averaging:** Aggregate independent zero-shot IRI predictions across N asynchronous vehicle passes per 100 m segment; spatial ensemble averaging suppresses zero-mean telemetry noise by $1/\sqrt{N}$.
- 3) **Phase 3 — Sensor Fusion and Drift Mitigation:** Future work will incorporate RTK GNSS carrier-phase positioning to mitigate longitudinal odometer drift [19], alongside real-time suspension damper temperature compensation and semi-supervised adaptation on unlabeled crowdsourced streams.

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